

Analysis

Jugo Dashboard

1. Introduction

Organizations use dashboards and charts to analyze business data. However, dashboards are often designed inconsistently, making them difficult to understand.

The **International Business Communication Standards (IBCS)** provide rules to standardize business reporting and data visualization.

One of these rules is **UN 3.2 – Unify Scenarios**, which defines how different types of data scenarios should be visually represented.

Examples of scenarios include:

- Actual (real measured data)
- Previous Year
- Plan
- Budget
- Forecast

These scenarios must be visually distinguishable in charts using **specific color and pattern rules**.

The goal of this project is to build an **AI-powered system that analyzes a dashboard image and checks whether it follows this IBCS rule**.

2. Project Objective

The main objective of this project is to develop an **AI application that can analyze data visualizations and detect whether they follow IBCS rule UN 3.2**.

The system should:

1. Analyze a dashboard or chart image
2. Detect scenario visualization elements
3. Determine if the chart follows IBCS standards
4. Provide feedback on how to improve the visualization

3. Scope of the Project

In Scope

The system should:

- Analyze **images of dashboards or charts**
- Detect **visual elements** (bars, lines, areas)
- Identify **scenario types** such as:
 - Actual
 - Previous Year
 - Plan
 - Budget
 - Forecast
- Check if the visualization follows **IBCS scenario notation**
- Provide feedback and improvement suggestions

Out of Scope

The system will **not**:

- Generate complete dashboards
- Modify dashboards automatically
- Analyze raw datasets

The system will only **analyze visualization images**.

4. Requirements

4.1 Functional Requirements

FR-01 (Must)

The system must allow a user to upload an image of a dashboard or data visualization.

FR-02 (Must)

The system must analyze the uploaded image to detect elements relevant to the IBCS *Unify Scenarios* rule.

FR-03 (Must)

The system must determine whether the uploaded visualization complies with the IBCS *Unify Scenarios* rule.

FR-04 (Must)

The system must provide feedback indicating whether the visualization is compliant or non-compliant.

FR-05 (Must)

The system must present the analysis results and feedback to the user in a clear interface.

FR-06 (Should)

The system should provide suggestions on how the visualization could be modified to improve IBCS compliance.

FR-07 (Should)

The system should provide visual feedback (e.g., highlighted elements or annotations) showing where compliance issues occur.

FR-08 (Must)

The user must be able to initiate the analysis of an uploaded dashboard image.

FR-09 (Could)

The user could delete uploaded images from the system.

FR-10 (Could)

The system could be designed to support scalability for additional IBCS rules in the future.

4.2 AI Recognition Requirements

AR-01 (Must)

The AI model must recognize scenario categories including **Actual, Previous Year, Plan, Budget, and Forecast**.

AR-02 (Must)

The model must detect scenario representation through **visual notation (fill styles, outlines, hatch patterns)** rather than relying solely on labels.

AR-03 (Must)

The model must recognize **Actual scenarios** represented by **solid dark fills** in visualization elements.

AR-04 (Must)

The model must recognize **earlier measured periods (e.g., Previous Year)** represented by **lighter solid fills**.

AR-05 (Must)

The model must recognize **planned scenarios (Plan or Budget)** represented by **outlined or bordered visualization elements**.

AR-06 (Must)

The model must recognize **forecast scenarios** represented by **outlined elements filled with hatch patterns**, where stripe colors correspond to measured data colors.

AR-07 (Should)

The model should detect scenario abbreviations such as **AC, PY, PL, BU, and FC** where present.

AR-08 (Should)

The model should recognize **variance charts where axes use scenario notation corresponding to the reference scenario**.

AR-09 (Must)

The model must detect scenario notation in **stacked or multi-series charts**, where notation is applied to the **lowest segment** and upper segments use **different gray shades**.

AR-10 (Should)

The model should recognize hatch patterns or outlines applied to stacked segments indicating planned or forecasted scenarios.

4.3 Data Requirements

DR-01 (Must)

The training dataset must contain both **IBCS-compliant and non-compliant dashboard images**.

DR-02 (Must)

The training dataset must contain at least **150 IBCS-compliant dashboard images**.

DR-03 (Must)

The training dataset must contain at least **150 non-IBCS-compliant dashboard images**.

DR-04 (Must)

Images in the dataset must be **labeled according to compliance with the IBCS Unify Scenarios rule**.

DR-05 (Should)

The dataset should contain dashboards featuring **different chart types**, including bars, columns, line charts, stacked charts, and multi-series charts.

DR-06 (Should)

The dataset should contain images from **different dashboard tools or styles** (e.g., Power BI, Tableau, Excel, SAP) to improve model generalization.

DR-07 (Must)

All training images must be **labeled consistently according to the same interpretation of the IBCS Unify Scenarios rule**.

DR-08 (Should)

The dataset should contain **varied dashboard layouts, chart styles, and image qualities** to ensure the model can generalize to different real-world dashboards.

4.4 Non-Functional / Quality Requirements

NFR-01 (Must)

The solution must be based on **AI techniques capable of analyzing dashboard images**.

NFR-02 (Should)

The model should achieve a **target accuracy of 70–80% when classifying IBCS compliance**.

NFR-03 (Could)

The system could incorporate **image recognition models combined with large language models** to generate feedback.

NFR-04 (Should)

The system should be designed in a way that allows **future extension to additional IBCS rules**.